BowlByte

# Intelligent canteen automation, smart queues, and food waste records

Team: [not recorded] · Guide: [not recorded] · College: [not confirmed in the repository]

## Problem

Counters guess readiness, can promise the last portion twice, and rarely keep an audit of cash or discarded food.

## Proposal

Students order in the browser. The kitchen works by station. Cashiers record cash and scan a one-time QR. Managers record production, waste, and a forecast they must approve.

## Architecture

Next.js and FastAPI, PostgreSQL, Redis, Celery. The browser stays on one origin. Money and stock change only inside database transactions.

## Modules

- Accounts and canteen roles
- Menu, reservation, kitchen ETA
- Cash receipts and QR pickup
- Unclaimed-order hold
- Forecast and preparation advice

## Evidence

Automated tests cover ordering, cash races, and one-time pickup. On a seeded series, seasonal naive validation MAE was 2.143 and was kept over gradient boosting at 3.000.

## Conclusion

Ready for a local demonstration and a supervised counter if a roster is supplied. Not a public site. Phone notifications were not observed. No waste reduction is claimed.

A1 portrait PDF: BOWLBYTE_POSTER.pdf (page size about 1684 by 2385 points). Editable source is this HTML file. Edge's default print was letter size and was replaced.